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United States Patent	12405408
Kind Code	B2
Date of Patent	September 02, 2025
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Optical film and display device

Abstract

An optical film and a display device are provided. The optical film includes: multiple optical structures, wherein one side of each optical structure away from a display surface is provided with multiple arced surfaces for reflecting at least a portion of environment light, and each optical structure has a first refractive index; and a filling part being filled between the optical structures and having a second refractive index greater than the first refractive index. Adjusting curvatures of the arced surfaces can implement the adjustment of the angle of reflection to the environment light.

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Appl. No.:	17/618510
Filed (or PCT Filed):	November 11, 2021
PCT No.:	PCT/CN2021/130144
PCT Pub. No.:	WO2023/070751
PCT Pub. Date:	May 04, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240036239 A1	Feb. 01, 2024

Foreign Application Priority Data

CN

202111283626.4

Nov. 01, 2021

Publication Classification

Int. Cl.: G02B5/02 (20060101); G02F1/1335 (20060101); H10K59/80 (20230101)

U.S. Cl.:

CPC G02B5/0231 (20130101); G02F1/133504 (20130101); H10K59/878 (20230201);

Field of Classification Search

CPC: G02B (5/0231); G02F (1/133504); G02F (1/1335); G02F (1/133553); H10K (59/878); H10K (59/879); H10K (59/10)

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Background/Summary

FIELD OF INVENTION

(1) This disclosure relates to a technical field of a display device, and more particularly to an optical film and a display device.

BACKGROUND OF INVENTION

(2) The reflectivity is an important optical specification of a large-size display device. The reflectivity represents the amount of light that can be reflected by the display device when a user watches the extinguished display device from a front side of the display device. Considering the user's habit of using the display device, the user usually does not wish to see the reflected light from the extinguished display device. That is, the display device is required to have the lower reflectivity to protect the user's privacy.

(3) However, the conventional optical film usually has the poor effect of decreasing the reflectivity.

SUMMARY OF INVENTION

Technical Problems

(4) In the prior art, the optical film usually has the poor effect of decreasing the reflectivity.

Technical Solutions

(5) An embodiment of this disclosure provides an optical film and a display device to solve the problem that the conventional optical film usually has the poor effect of decreasing the reflectivity.

(6) In a first aspect, the embodiment of this disclosure provides an optical film applied to a display panel. The optical film is disposed on one side of a display surface of the display panel. The optical film includes: multiple optical structures, wherein one side of each of the optical structures away from the display surface is provided with multiple arced surfaces for reflecting at least a portion of environment light, and each of the optical structures has a first refractive index; and a filling part, which is disposed on one side of the optical structures away from the display surface, covers the optical structures, is filled between the optical structures, and has a second refractive index greater than the first refractive index.

(7) An embodiment of this disclosure further provides a display device including a display panel and the optical film mentioned hereinabove.

Useful Effects

(8) In the optical film and the display device according to the embodiments of this disclosure, it is possible to adjust values of refractive indexes of the filling part and the optical structure to adjust a critical angle of refraction between the optical structure and the filling part, thereby implementing the total internal reflection (TIR) of the optical structure. In addition, arced surfaces capable of reflecting at least a portion of environment light are provided in the optical structures, so that the curvatures of the arced surfaces can be adjusted to implement the adjustment of the angle of reflection to the environment light, so that the requirement of the low reflectivity is implemented through the optical film in the display device.

Description

DESCRIPTION OF DRAWINGS

(1) In order to illustrate the technical solutions in the embodiments of this disclosure more clearly, the drawings required for the descriptions of the embodiments will be briefly introduced in the following. Obviously, the drawings in the following description are just some embodiments of this

disclosure. For those skilled in the art, other drawings can be obtained from these drawings without creative works. In order to make this disclosure and the beneficial effects thereof be understood more completely, explanations will be made in the following in conjunction with the drawings. In the following description, the same reference numerals refer to the same parts.

(2) FIG. 1 is a schematic structure view showing a display device provided by an embodiment of this disclosure.

(3) FIG. 2 is a schematic structure view showing an optical film in the display device shown in FIG. 1.

(4) FIG. 3 is a first schematic cross-sectional structure view showing the optical film of FIG. 2 taken in the A-A direction.

(5) FIG. 4 is a second schematic cross-sectional structure view showing the optical film of FIG. 2 taken in the A-A direction.

(6) FIG. 5 is a third schematic cross-sectional structure view showing the optical film of FIG. 2 taken in the A-A direction.

(7) FIG. 6 is a schematic flow chart showing a preparation method of an optical film provided by an embodiment of this disclosure.

DETAILED DESCRIPTION OF EMBODIMENTS

(8) The technical solutions in the embodiments of this disclosure will be clearly and completely described in the following with reference to the drawings of the embodiments of this disclosure. Obviously, the described embodiments are only a part of the embodiments of this disclosure, rather than all the embodiments. Based on the embodiments of this disclosure, all other embodiments obtained by those skilled in the art without creative works are deemed as falling within the scope of this disclosure.

(9) The reflectivity is an important optical specification in large-size display devices, such as a liquid crystal display (LCD), an organic light-emitting diode (OLED) display or an organic light-emitting semiconductor display, and the like. The reflectivity represents the amount of light that can be reflected by the display device when the user watches the extinguished display device from a front side of the display device. Considering the user's habit of using the display device, the user usually does not wish to see the reflected light from the extinguished display device. That is, the display device is required to have the lower reflectivity. The low reflectivity usually means that the less amount of light that can be reflected by the display device in the extinguished state of the display device, so that the user cannot implement the operation of "looking in the mirror" through the screen of the display device, the user's privacy can be protected, and the display device has the better performance.

(10) However, the conventional optical film usually has the poor effect of decreasing the reflectivity.

(11) In order to solve the above-mentioned problems, the embodiments of this disclosure provide an optical film and a display device, and the following explanations will be made in conjunction with the drawings.

(12) Exemplarily, please refer to FIG. 1. FIG. 1 is a schematic structure view showing a display device provided by an embodiment of this disclosure. The embodiment of this disclosure provides a display device 1, which may be a device, such as a mobile phone, a desktop computer, a notebook, a tablet computer, a wrist watch apparatus, a television, a media playback apparatus, a computer monitor and the like, having a display function. The display device 1 may include a display panel 10 and an optical film 20. The display panel 10 can output light. The display panel 10 has a display surface and a non-display surface. The display surface may be understood as one side of the display panel 10 displaying output light, and the non-display surface is the other side disposed opposite the display surface and does not output light. The optical film 20 of the embodiment of this disclosure can improve the problem of the display device 1 having the low reflectivity in the extinguished state to cause the poor effect, and the optical film 20 can adjust an

exit angle of output light of the display panel **10**.

(13) In order to explain the principle of action and structural compositions of the optical film **20** of the embodiment of this disclosure more clearly, explanations to the structural compositions and the principle of action of the optical film **20** will be made in the following in conjunction with the drawings.

(14) Exemplarily, please refer to FIGS. **2** and **3** in conjunction with FIG. **1**. FIG. **2** is a schematic structure view showing an optical film in the display device shown in FIG. **1**. FIG. **3** is a first schematic cross-sectional structure view showing the optical film of FIG. **2** taken in the A-A direction. The embodiment of this disclosure provides the optical film **20** applied to the display panel **10**. The optical film **20** is disposed on one side of the display surface of the display panel **10**. The optical film **20** may include multiple optical structures **21**. One side of each optical structure **21** away from the display surface is provided with multiple arced surfaces **2142** for reflecting at least a portion of environment light. Each optical structure **21** has a first refractive index. Each optical film **20** further includes a filling part **23**, which is disposed on one side of the optical structures **21** away from the display surface, covers the optical structures **21**, is filled between the optical structures **21**, and has a second refractive index greater than the first refractive index.

(15) In the optical film **20** of the embodiment of this disclosure, adjusting values of the refractive indexes of the filling part **23** and the optical structure **21** can adjust a critical angle of refraction between the optical structure **21** and the filling part **23**, and thus implement the TIR of the optical structure **21**. Also, the optical structures **21** are provided with the arced surfaces **2142** capable of reflecting the at least a portion of environment light, so that curvatures of the arced surfaces **2142** can be adjusted to implement the adjustment of the angle of reflection to the environment light, and that the requirement of the low reflectivity can be implemented through the optical film **20** in the display device **1**.

(16) It is to be described that the optical film **20** needs to satisfy the requirement of TIR in order to implement the requirement of the low reflectivity of the optical film **20**. TIR is an optical phenomenon. When the light enters the medium having the lower refractive index from the medium having the higher refractive index, the refracted light will disappear if an incident angle is greater than a certain critical angle (the light travels away from the normal), and all incident light rays are reflected without entering the medium having the low refractive index. Therefore, the second refractive index of the filling part **23** is configured to be greater than the first refractive index of the optical structure **21**, thereby implementing the TIR of the environment light to the optical structure **21** through the filling part **23**.

(17) TIR is a first step for the implementation of the low reflectivity of the optical film **20**. The low reflectivity of the optical film **20** means that a less portion of light of TIR is reflected to one side of the display surface, and may also be understood as that the external environment light reaches the surface of the optical structure **21** through the filling part **23**, and then reflected to the non-display surface. Thus, the user cannot see the reflection of the environment light when watching the display surface. Based on this, the requirement of the low reflectivity of the optical film **20** can be implemented. In order to implement the above-mentioned optical reflection, the embodiment of this disclosure adjusts the reflected light by providing the arced surfaces **2142**, so that reflected light can be reflected to the non-display surface by adjusting the curvatures of the arced surfaces **2142**.

(18) The optical film **20** of the embodiment of this disclosure can further adjust the exit angle of the output light of the display panel **10** to implement the requirement of the wide viewing angle of the optical film **20**.

(19) It is to be described that the brightness performance of the large-size display device gets worse and worse as the viewing angle becomes larger. To solve this problem, the prior art usually uses the optical film to improve the problem of the large viewing angle. The objective of improving the performance of the display device at the large viewing angle can be achieved by additionally providing films, having different micro-structures, on the display device. Meanwhile, in order to

further satisfy the lower external light reflectivity of the surface of the display device, a polarizer or an anti-reflection film needs to be attached to the display panel to decrease the surface reflectivity. Therefore, films having two or more than two functions need to be combined in order to satisfy the specifications of the wide viewing angle and the low reflectivity of the display device.

(20) However, after multiple films are combined, the optical property of the composite film changes as compared with that of one single film, and whether the optical performance thereof satisfies the requirement needs to be evaluated again. In addition, the processes of the composite film also become more complicated. If the composite film is used in the mass production, the manufacturing cost of the film is increased, and the corresponding composite process needs to be added, thereby increasing the production time and the labor cost. Therefore, a new film needs to be developed to improve the above-mentioned condition.

(21) Exemplarily, each optical structure **21** may include a bottom wall **212** and multiple sidewalls **214**. The bottom wall **212** is disposed on one side of the sidewalls **214** facing the display panel **10**. The sidewalls **214** may tilt to the bottom wall **212**. The arced surfaces **2142** are disposed on at least one sidewall **214**. At least a portion of light outputted from the display panel **10** is incident to the optical structure **21** from the bottom wall **212** at a first incident angle, and is refracted out of the optical structure **21** through the arced surfaces **2142** at a first refraction angle. The first refraction angle is greater than the first incident angle. At least a portion of environment light is reflected to the non-display surface of the display panel **10** through the arced surfaces **2142**.

(22) In the optical film **20** of the embodiment of this disclosure, the sidewalls **214** of the optical structure **21** tilt to the bottom wall **212**, so that the first refraction angle of the output light of the display panel **10** can be as large as possible by adjusting the tilt angles, thereby satisfying the requirement of the wide viewing angle of the optical film **20**. The optical film **20** of the embodiment of this disclosure can satisfy the requirement of the wide viewing angle as well as the requirement of the low reflectivity. It is only necessary to configure the optical structure **21** to have the prismatic shape and to dispose the arced surfaces **2142** on the optical structure **21** to implement both of the above-mentioned requirements. Compared with the conventional composite film, the process becomes simple.

(23) It is to be described that the first refraction angle can be adjusted by adjusting the tilt angle of the sidewall **214** relatively to the bottom wall **212** and the height of the optical structure **21** so that the optical film **20** has the larger refraction angle, that the user can watch the display device **1** from various angles to obtain the same display effect, and that the requirement of the wide viewing angle of the display device **1** is satisfied. The optical structure **21** may further make the environment light be reflected to the non-display surface of the display panel **10** through the arced surfaces **2142**, but not make the environment light be reflected to the incident direction of the environment light, so that the reflectivity of the display device **1** can be decreased, and the requirement of the low reflectivity of the display device **1** to the environment light is satisfied. The embodiment of this disclosure can concurrently satisfy the requirements of the wide viewing angle and the low reflectivity of the display device **1** using one optical film **20**. Compared with the conventional composite film, the process becomes simple.

(24) Please refer to FIG. 4 in conjunction with FIGS. 1 to 3. FIG. 4 is a second schematic cross-sectional structure view showing the optical film of FIG. 2 taken in the A-A direction. Each arced surface **2142** may be depressed toward the bottom wall **212**, so that the optical structure **21** is provided with the structures of multiple depressed arced surfaces **2142**. In some embodiments, please refer to FIG. 5 in conjunction with FIGS. 1 and 2. FIG. 5 is a third schematic cross-sectional structure view showing the optical film of FIG. 2 taken in the A-A direction. Each arced surface **2142** may further project away from the bottom wall **212**, so that the optical structure **21** is provided with the structures of multiple convex arced surfaces **2142**. It is to be described that the provision of the arced surfaces **2142** can change the direction of normal of the light, thereby implementing the requirement of the low reflectivity of the optical film **20**. Therefore, either the

depressions or projections of the arced surfaces **2142** are provided with respect to the bottom wall **212**, the optical structure **21** of the embodiment of this disclosure can change the direction of normal of the environment light, thereby adjusting the angle of reflection of the environment light, and implementing the requirement of the low reflectivity of the optical film **20**.

(25) Exemplarily, the arced surfaces **2142** may have the same arc curvatures. Such the arced surfaces **2142** can make the optical structures **21** be conveniently manufactured. Exemplarily, the radius of curvature may be an arbitrary value ranging from 2 microns to 4 microns. For example, the radius of curvature of each of the arced surfaces **2142** may be equal to 2.5, 2.8 or 3 microns. By selecting the arc curvatures of the arced surfaces **2142**, it is possible to adjust the angle of reflection of the environment light to implement the requirement of the low reflectivity of the optical film **20**.

(26) Exemplarily, two sidewalls **214** extending in a lengthwise direction of the optical structure **21** are respectively provided with multiple arced surfaces **2142**. Inner angles respectively formed between the two sidewalls **214** and the bottom wall **212** are equal to each other. It is to be described that the optical structure **21** can extend from an end portion of the display panel **10** in the lengthwise or widthwise direction of the display panel **10**. This extending direction is the lengthwise direction of the optical structure **21**. The two sidewalls **214** are respectively provided with the arced surfaces **2142** in the lengthwise direction of the optical structure **21**. It is understandable that the optical structure **21** may include a bottom wall **212** and four sidewalls **214**, wherein two of the four sidewalls **214** extending in the lengthwise direction of the optical structure **21** are disposed opposite each other, and multiple arced surfaces **2142** are provided on each sidewall **214**. The two sidewalls **214** provided with the arced surfaces **2142** are disposed opposite each other. Included angles between the bottom wall **212** and the sidewalls **214** are equal to each other. That is, inner angles formed between the bottom wall **212** and the sidewalls **214** are equal to each other. The output light of the display panel **10** can be adjusted by adjusting the inner angles between the bottom wall **212** and the sidewalls **214** and the height of the optical structure **21**, so that the output light of the display panel **10** travelling through the optical film **20** has the first refraction angle, thereby implementing the requirement of the wide viewing angle of the optical film **20**. The two sidewalls **214** provided with the arced surfaces **2142** are disposed symmetrically. The other two sidewalls **214** in the four sidewalls **214** are disposed opposite each other in the lengthwise direction of the optical film **20**, and the two sidewalls **214** are not provided with the arced surface **2142**. The two sidewalls **214** without the arced surface **2142** may be flush with the end surface of the display panel **10**. In addition, the areas of the sidewalls **214** without the arced surface **2142** are smaller than the areas of the sidewalls **214** provided with the arced surfaces **2142**.

(27) The optical structure **21** may have a prismatic shape. For example, as shown in FIG. 3, the optical structure **21** may have a triangular prism shape. Ends of the two sidewalls **214** provided with the arced surfaces **2142** away from the bottom wall **212** are connected together. It is understandable that the two sidewalls **214** without the arced surface **2142** are triangular in the lengthwise direction of the optical structure **21**, and that two edges thereof have multiple arcs. For example, as shown in FIGS. 4 and 5, the optical structure **21** may further have a quadrangular prism shape. The optical structure **21** may further include a top wall **216** disposed opposite the bottom wall **212**, wherein two ends of the top wall **216** are respectively connected to the two sidewalls **214** provided with the arced surfaces **2142**. It is understandable that the two sidewalls **214** without the arced surface **2142** are trapezoidal in the lengthwise direction of the optical structure **21**, and two lateral sides of the trapezoid have multiple arcs.

(28) Exemplarily, referring again to FIGS. 1 to 3, the filling part **23** is disposed on one side of the optical structures **21** away from the display surface, covers the optical structures **21**, and is filled between the optical structures **21**. It is to be described that the optical structures **21** may be disposed in parallel with one side of the display surface of the display panel **10**. That is, the optical structures **21** are parallelly and sequentially arranged on one side of the display surface of the display panel **10**. A predetermined distance may be present between adjacent two optical structures

21. The adjacent two optical structures **21** may also abut upon each other. The embodiment of this disclosure is described according to the predetermined distance present between the adjacent two optical structures **21**. Multiple sequentially arranged optical structures **21** may be arranged in the lengthwise direction of the display panel **10**, and may also be arranged in the widthwise direction of the display panel **10**, and no restriction is made thereto. The filling part **23** covers the optical structures **21**, and is filled between the optical structures **21**. It is understandable that the filling part **23** covers the optical structures **21** and is filled between multiple optical structures **21** when adjacent two optical structures **21** abut upon each other. At this time, the bottommost end of the filling part **23** in the direction of the display surface and the bottom wall **212** of the optical structure **21** are not disposed on the same plane. When the predetermined distance is present between adjacent two optical structures **21**, the filling part **23** covers the optical structures **21**, and the filling part **23** and the bottom wall **212** of the optical structure **21** are disposed on the same plane. The filling part **23** and the optical structures **21** constitute a smooth film structure.

(29) Exemplarily, the optical film **20** may further include a base layer **25** disposed on one side of the optical structures **21** facing the display panel **10**. It is understandable that the optical film **20** may be a layered structure including the base layer **25**, the optical structures **21** and the filling part **23**. The base layer **25** is disposed at bottom, the optical structures **21** are disposed in parallel with the base layer **25**, and the filling part **23** covers the optical structures **21** and is filled between the optical structures **21**.

(30) The optical film **20** may be constituted by three materials. That is, the base layer **25**, the filling part **23** and the optical structure **21** are respectively formed using different materials. Exemplarily, the base layer **25** may have a third refractive index, which may be greater than the first refractive index and smaller than the second refractive index. That is, the second refractive index of the filling part **23** is greater than the third refractive index of the base layer **25**, and the third refractive index of the base layer **25** is greater than the first refractive index of the optical structure **21**. It is to be described that the light needs to travel from the optically dense medium to the optically less dense medium in order to implement the TIR of the environment light. Therefore, the second refractive index of the filling part **23** is greater than the first refractive index of the optical structure **21**. The output light of the display panel **10** needs to travel to the optical structure **21** through the base layer **25** and to be refracted out, and then refracted to the outside from the optical structure **21** through the filling part **23**, and the first refraction angle of light refracted to the outside needs to be as large as possible. When the light enters the medium, in which the speed of light is low (the refractive index is low) from the medium, in which the speed of light is high (the refractive index is high), the refraction angle is smaller than the incident angle. When the light enters the medium, in which the speed of light is high, from the medium, in which the speed of light is low, the refraction angle is greater than the incident angle. Therefore, it is possible to configure the second refractive index of the filling part **23** to be greater than the third refractive index of the base layer **25**, and then to configure the third refractive index of the base layer **25** to be greater than the first refractive index of the optical structure **21**, so that the output light of the display panel **10** is refracted to the optical structure **21** at the smaller refraction angle, and then refracted out of the optical film **20** at the larger first refraction angle, thereby implementing the requirement of the wide viewing angle of the optical film **20**. Exemplarily, the materials of the optical structures **21**, the filling part **23** and the base layer **25** may be resin materials having three different refractive indexes. For example, the refractive index of the resin material may range from 1.4 to 1.7, so the first refractive index of the optical structure **21**, the second refractive index of the filling part **23** and the third refractive index of the base layer **25** may be respectively equal to 1.4, 1.6 and 1.7, and that the optical film **20** can satisfy the requirement of the wide viewing angle and further satisfy the requirement of the low reflectivity.

(31) It is to be described that the adjustment of the output light can be implemented by adjusting the inner angle between the bottom wall **212** of the optical structure **21** and the sidewall **214** of the

optical structure **21** provided with the arced surfaces **2142**, adjusting the height of the optical structure **21**, and selecting the materials of the base layer **25**, the filling part **23** and the optical structures **21** to have different refractive indexes. Thus, the optical structure **21** can be used to diverge the incident light with the small viewing angle to the output light with the large viewing angle and thus to improve the viewing angle or achieve the objective of the wide viewing angle. In addition, the TIR of the arced surface **2142** can be implemented by adjusting the arc curvatures of the structures of the arced surfaces **2142** in conjunction with selecting of the refractive indexes of the base layer **25**, the filling part **23** and the optical structures **21**, thereby decreasing the surface light reflectivity of the display device **1**.

(32) Exemplarily, the inner angle between the sidewall **214** and the bottom wall **212** may range from 50° to 80°, and the height of the optical structure **21** may range from 10 microns to 30 microns. For example, parts of the optical film **20** of the embodiment of this disclosure may have the following configuration parameters. The inner angle between the sidewall **214** and the bottom wall **212** is equal to 60°. The height of the optical structure **21** may be equal to 20 microns. The first refractive index of the optical structure **21**, the second refractive index of the filling part **23** and the third refractive index of the base layer **25** may be respectively equal to 1.4, 1.6 and 1.7. The radius of curvature of the structure of the arced surface **2142** may be equal to 2.5 microns. The above-mentioned parameter configurations of the parts of the optical film **20** can satisfy the requirements of the low reflectivity and the wide viewing angle of the optical film **20**. Of course, the parameter configurations of the parts of the optical film **20** are not restricted to the above-mentioned conditions, and detailed examples thereof will be omitted herein.

(33) It is to be described that the configurations of the optical film **20** of the embodiment of this disclosure can increase the viewing angle of the display device **1** while decreasing the surface reflectivity of the display device **1**. It is unnecessary to combine various functions of films, so that the process is simple and adapted to the mass production.

(34) Exemplarily, please refer to FIG. **6**. FIG. **6** is a schematic flow chart showing a preparation method of an optical film provided by an embodiment of this disclosure. The embodiment of this disclosure further provides a preparation method of an optical film. The structure and the principle of action of the optical film **20** may be found in the above-mentioned explanations corresponding to FIGS. **1** to **5**, and detailed descriptions thereof will be omitted herein. The optical film **20** is applied to the display panel **10**, and disposed on one side of the display surface of the display panel **10**. The preparation method of the optical film includes steps **101** to **103**.

(35) In the step **101**, a substrate is provided.

(36) Exemplarily, before the optical structure **21** is manufactured, a smooth substrate on which multiple optical structures **21** are placed is needed. During production, the optical structures **21** may be directly formed on the substrate. After the optical structure **21** is formed, the substrate is etched off to obtain the optical structures **21**. It is also possible to form a base layer **25** on the substrate in advance, and then to form the multiple optical structures **21**. At last, the formed base layer **25** and the optical structures **21** are removed from the substrate.

(37) In the step **102**, multiple optical structures are formed on the substrate. One side of each optical structure away from the display surface is provided with multiple arced surfaces for reflecting at least a portion of environment light. Each optical structure has a first refractive index.

(38) Exemplarily, a base material layer having multiple optical structures **21** arranged at equal intervals or adjacently arranged may be firstly formed on the substrate. For example, the base material layer may present a triangular prism shape, and the bottom wall of the base material layer contacts the substrate. Thereafter, a formation process is used to form the structures of the arced surfaces **2142** on the base material layer and thus to form the optical structures **21**. The formation process may be a roll to roll process. Of course, the formation process may also be other thermoforming processes. For example, the optical structure **21** may also be formed in the package material by directly adopting the lithographing process (e.g., thin film encapsulation (TFE)), and

may be directly used after being manufactured.

(39) In the step **103**, a filling part is formed on one side of the optical structures away from the display surface, wherein the filling part covers the optical structures, is filled between the optical structures, and has a second refractive index greater than the first refractive index.

(40) After the optical structures **21** have been formed, a filling material may be injected onto the outsides of the optical structures **21** to form a smooth outer surface, and thus to form the structure of the filling part **23**. The filling part **23** covers the optical structures **21** and is filled between the optical structures **21**.

(41) In some embodiments, when the base layer **25** is disposed on the substrate, the base layer **25**, the optical structures **21** and the filling part **23** after being formed commonly constitute the optical film **20**.

(42) The optical film **20** of the embodiment of this disclosure may be disposed on the light emitting side of the display panel **10** and used. At least a portion of light outputted from the display panel **10** is incident to the optical structure **21** from the bottom wall **212** at a first incident angle, and is refracted out of the optical structure **21** through the arced surface **2142** at a first refraction angle. The first refraction angle is greater than the first incident angle. At least a portion of environment light is reflected to the non-display surface of the display panel **10** through the arced surfaces **2142**. In the optical film **20** of the embodiment of this disclosure, adjusting the values of the refractive indexes of the filling part **23** and the optical structure **21** can adjust the critical angle of refraction between the optical structure **21** and the filling part **23**, and thus implement the TIR of the optical structure **21**. In addition, the arced surfaces **2142** capable of reflecting at least a portion of environment light are disposed in the optical structures **21**, so that the curvatures of the arced surfaces **2142** can be adjusted to implement the adjustment of the angle of reflection to the environment light and to implement the adjustment of the angle of reflection to the environment light, and that the display device **1** can implement the requirement of the low reflectivity through the optical film **20**. The first refraction angle of the output light of the display panel **10** can be as large as possible by adjusting the tilt angles, thereby satisfying the requirement of the wide viewing angle of the optical film **20**. The optical film **20** of the embodiment of this disclosure can satisfy the requirement of the wide viewing angle as well as the requirement of the low reflectivity. It is only necessary to configure the optical structure **21** to have the prismatic shape, and to configure the arced surfaces **2142** on the optical structure **21**, so that both the above-mentioned requirements can be satisfied. Compared with the conventional composite film, the process becomes simple.

(43) In the above-mentioned embodiments, the description of each embodiment has its own focus. For parts that are not described in detail in one embodiment, reference can be made to the relevant description of other embodiments.

(44) In the description of this disclosure, the terms “first” and “second” are only used for descriptive purposes, and cannot be understood as indicating or implying relative importance or implicitly indicating the number of technical features as indicated. Thus, the features restricted by “first” and “second” may expressly or implicitly include one or more features.

(45) Detailed introductions have been made to the optical film and the display device provided by the embodiments of this disclosure. In this disclosure, specific examples are used to explain the principles and implementation of this disclosure. The descriptions of the above-mentioned embodiments are only used to help the understanding of the method and core idea of this disclosure. Meanwhile, those skilled in the art will make changes to the specific implementation and the application scope according to the idea of this disclosure. In summary, the contents of this specification should not be construed as a limitation to this disclosure.

Claims

1. An optical film applied to a display panel, wherein the optical film is disposed on one side of a display surface of the display panel, and the optical film comprises: multiple optical structures, wherein one side of each of the optical structures away from the display surface is provided with multiple arced surfaces for reflecting at least a portion of environment light, and each of the optical structures has a first refractive index; a filling part, which is disposed on one side of the optical structures away from the display surface, covers the optical structures, is filled between the optical structures, and has a second refractive index greater than the first refractive index; and a base layer disposed on one side of the optical structures facing the display panel; wherein each of the optical structures is in contact with the base layer; each of the optical structures comprises a bottom wall and multiple sidewalls, wherein the bottom wall is disposed on one side of the sidewall facing the display panel, and the sidewalls tilt to the bottom wall; the base layer has a third refractive index, which is greater than the first refractive index and smaller than the second refractive index.
2. The optical film according to claim 1, wherein the arced surfaces are disposed on at least one of the sidewalls.
3. The optical film according to claim 2, wherein at least a portion of light outputted from the display panel is incident to the optical structures from the bottom wall at a first incident angle, and is refracted out of the optical structures through the sidewall at a first refraction angle greater than the first incident angle; and at least a portion of environment light is reflected to a non-display surface of the display panel through the arced surfaces.
4. The optical film according to claim 2, wherein each of the arced surfaces is depressed toward the bottom wall or projects away from the bottom wall.
5. The optical film according to claim 2, wherein two of the sidewalls extending in a lengthwise direction of the optical structure are respectively provided with multiple ones of the arced surfaces, and inner angles respectively formed between two of the sidewalls and the bottom wall are equal to each other.
6. The optical film according to claim 5, wherein ends of two of the sidewalls away from the bottom wall are connected together.
7. The optical film according to claim 5, wherein each of the optical structures further comprises a top wall, which is disposed opposite the bottom wall and connected to two of the sidewalls.
8. A display device, comprising a display panel and an optical film, wherein the optical film is disposed on one side of a display surface of the display panel, and the optical film comprises: multiple optical structures, wherein one side of each of the optical structures away from the display surface is provided with multiple arced surfaces for reflecting at least a portion of environment light, and each of the optical structures has a first refractive index; a filling part, which is disposed on one side of the optical structures away from the display surface, covers the optical structures, is filled between the optical structures, and has a second refractive index greater than the first refractive index; and a base layer disposed on one side of the optical structures facing the display panel; wherein each of the optical structures is in contact with the base layer; each of the optical structures comprises a bottom wall and multiple sidewalls, wherein the bottom wall is disposed on one side of the sidewall facing the display panel, and the sidewalls tilt to the bottom wall; the base layer has a third refractive index, which is greater than the first refractive index and smaller than the second refractive index.
9. The display device according to claim 8, wherein the arced surfaces are disposed on at least one of the sidewalls.
10. The display device according to claim 9, wherein at least a portion of light outputted from the display panel is incident to the optical structures from the bottom wall at a first incident angle, and is refracted out of the optical structures through the sidewall at a first refraction angle greater than the first incident angle; and at least a portion of environment light is reflected to a non-display surface of the display panel through the arced surfaces.

11. The display device according to claim 9, wherein each of the arced surfaces is depressed toward the bottom wall or projects away from the bottom wall.
 12. The display device according to claim 9, wherein two of the sidewalls extending in a lengthwise direction of the optical structure are respectively provided with multiple ones of the arced surfaces, and inner angles respectively formed between two of the sidewalls and the bottom wall are equal to each other.
 13. The display device according to claim 12, wherein ends of two of the sidewalls away from the bottom wall are connected together.
 14. The display device according to claim 12, wherein each of the optical structures further comprises a top wall, which is disposed opposite the bottom wall and connected to two of the sidewalls.
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